SPGC: Shape-Prior-Based Generated Content Data Augmentation for Remote Sensing Object Detection

Yalun Dai, Fei Ma, *Member, IEEE*, Wei Hu, and Fan Zhang, *Senior Member, IEEE*

***Abstract*— While deep learning-based methods have made significant strides in remote sensing applications, the scarcity and inadequate quality of remote sensing images tend to curtail the improvement of follow-up research such as remote sensing object detection. However, the human visual system is able to quickly grasp the features of an unseen object given only a few examples, which is considered to be related to a strong shape bias. Inspired by how human toddlers learn shapes and the process of recognizing objects by shape, this article proposes a novel method known as shape-prior-based generated content (SPGC) data augmentation to overcome these challenges. Specifically, our method includes two main steps: shape data generation and stylization. Initially, the method begins with generating shape data regardless of training data availability. Next, we enhance the robustness of the generated shape data through stylization, forming a robust shape dataset (SD). Stylization is further bifurcated into two scenarios: when training data are unseen, self-stylization is employed where the shape data simultaneously serves as content and style data, resulting in significant performance improvements. When training data are accessible, data-specific stylization is applied, with the shape data as content and training data as style, leading to more substantial enhancements than self-stylization. Experimental results on mainstream remote sensing object detection datasets including NWPU VHR-10, DIOR, and FAIR1M demonstrate that our method significantly improves performance and underscores its effectiveness.**

***Index Terms*— Data augmentation, remote sensing object detection, shape-prior guidance, stylization.**

I. Introduction

OBJECT detection in optical remote sensing images is a fundamental and difficult task in the field of aerial and satellite image analysis. This pivotal task is used in a wide range of applications, such as land planning, urban monitoring, and traffic planning, and has attracted increasing interest in recent years. Thanks to its substantial enhancement of a variety of image processing techniques, deep learning has emerged as a powerful tool to tackle this issue [1], [2], [3], [4]. And deep learning methods applied to object detection have garnered intense research interest, setting new standards for the state-of-the-art performance in the field.

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The authors are with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: dialogue_dylan@outlook.com; zhangf@mail.buct.edu.cn).

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While deep learning-based object detection methods are powerful, their inherent limitations stem primarily from heavy reliance on vast quantities of high-quality annotated data. This reliance becomes particularly challenging in remote sensing imagery due to its unique data acquisition issues and variable image quality. One significant issue is the collection of large and comprehensive datasets, which is often a difficult and sometimes impossible task in remote sensing, leading to a scarcity of data for model training. Additionally, the varying quality of these images introduces additional complexity to the training process, potentially resulting in models that generalize poorly to unseen data. To tackle these problems, researchers have turned to data augmentation techniques. These methods create additional training data by modifying existing images in various ways, effectively increasing the size and diversity of the training dataset and enhancing the model's robustness.

In the realm of remote sensing, data augmentation plays a crucial role in improving image analysis and interpretation, utilizing three core approaches: global transformation, element synthesis, and conditional generation. Global transformation enhances image quality and diversifies datasets through basic operations such as random transformation, image erasing [5], and mixing [6], [7], [8], [9]. In contrast, element synthesis [10] uses an instance-based approach to manipulate and recombine image elements, thereby enriching the datasets in both quantity and quality. Conditional generation methods [11], [12], [13], [14] employ deep generative models to direct image generation under semantic constraints, ensuring key semantic information is embedded in the resulting images.

While these data augmentation techniques have significantly advanced remote sensing, they are accompanied by notable challenges. Global transformation methods, such as BoxPaste [8], effectively enhance object density through techniques like cropping and pasting, and are instrumental in expanding datasets. While this method is cost-effective and successfully expands the dataset size, it still presents challenges: *the potential to lead to overfitting, especially if the dataset size remains inadequate.* This issue stems from their reliance on basic transformations of the original data, failing to adequately diversify data distribution and patterns. In contrast, conditional generation methods, which include the latest techniques based on generative adversarial networks (GANs) [11] and denoising diffusion implicit models (DDIMs) [14], are renowned for producing realistic and semantically rich images. Despite their benefits, these methods face their own set of challenges,

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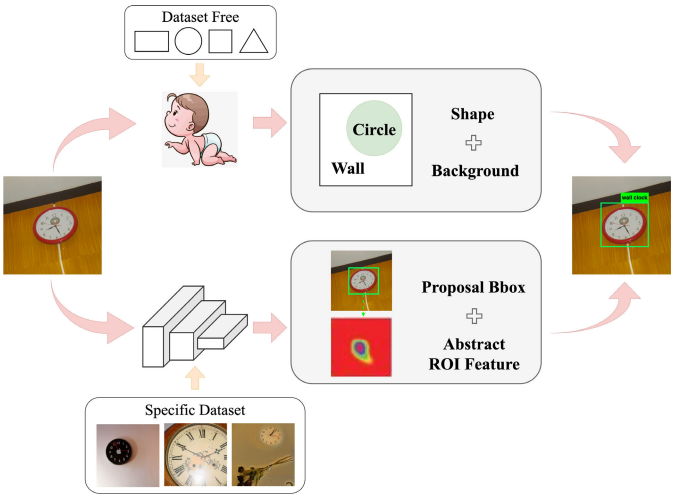


Fig. 1. Comparison of children's learning and reasoning processes with those of deep neural networks (DNNs). Children typically begin by learning shapes and subsequently recognize objects based on these forms. In contrast, DNNs directly learn from target data. While humans can identify new objects from a few examples, DNNs require extensive training data. Our goal is to develop the DNNs that learn and decide in a manner akin to humans, aiming to bridge the gap between deep learning and human vision.

including *the need for substantial training data and high computational resources*. The scarcity of remote sensing data further exacerbates the limitations in the quality of images generated by these methods.

By introducing shape priors into the model, our method adeptly addresses the aforementioned challenges. On the one hand, our method augments the dataset size by generating a shape dataset (SD), introducing a data distribution distinct from the original, thereby mitigating the overfitting issue. On the other hand, the generation of the SD not only entails minimal computational overhead but also provides a generative data augmentation strategy that necessitates no data from the target dataset.

Human vision shows a strong shape bias [15], [16]. This bias has historical roots, as evidenced by the use of pictographs, which underscore the importance of shape as a distinguishing feature among various objects. Such a feature remains consistent for similar objects. The human visual system is able to quickly grasp the feature of an unseen object given only a few examples, which is considered to be related to a strong shape bias [17]. However, this is in contrast to the current deep learning-based detection models, which heavily rely on vast and diverse datasets. They often struggle to extract meaningful knowledge from a limited number of samples. As shown in Fig. 1, some previous works [18], [19], [20] show that the strong shape bias in human vision is widely considered to come from early childhood shape learning and the recognition of objects through shapes. In response to these insights, motivated by the superior shape bias in the human visual system, we propose a shape-prior-based generated content (SPGC) data augmentation technique to improve object detection performance in remote sensing. By emulating the shape learning process during early human development, our method artfully introduces synthetic shape images with low computational overhead. As shown in Fig. 2, our method explicitly incorporates shape information into the object detection model, guiding its shape priors, and enhancing its ability to accurately locate objects in remote sensing images. Consequently, the detection models' performance is significantly improved. To be specific, our method involves generating shape data at first, a process that does not rely on any existing dataset. Following this, we incorporate a stylization procedure to enhance the robustness of the shape data. This stylization takes two forms: in the absence of available training data, we employ self-stylization where the shape data acts as both content and style, resulting in a significant performance boost. Conversely, when training data are available, we use dataset-specific stylization, with the shape data serving as content and the training data as style, leading to even greater performance enhancements than self-stylization. The philosophy behind our approach lies in exploiting the insights from the human visual system's shape bias and the inherent shape-invariance properties of the remote sensing targets, thereby making our data augmentation method more robust and efficient.

In general, our main contributions are summarized as follows.

1) We propose an innovative SPGC data augmentation method, inspired by the learning process of human toddlers recognizing shapes. This method explicitly utilizes shape priors to guide object detection.
2) The SPGC method generates shape data at a very low computational cost. This process effectively integrates shape priors into the object detection model, thereby enhancing its ability to accurately localize objects.
3) A stylization process is proposed to bolsters the robustness of models trained on shape data. This enhancement enables these models to generalize more effectively across diverse remote sensing datasets.
4) Our data augmentation approach is designed for seamless integration with mainstream object detection frameworks, including both one-stage and two-stage architectures, as well as anchor-based and anchor-free systems. Its efficacy and versatility are validated through comprehensive experiments conducted on the NWPU VHR-10 [1], DIOR [2], and FAIR1M [21] datasets, encompassing both CNN-based and transformer-based detectors.

The remainder of this article is organized as follows. In Section II, we briefly review related works on geospatial object detection algorithms, and data augmentation for Remote sensing images. In Section III, we describe the proposed method in detail. The details of our experiments and the results are presented in Section IV. Finally, Section V concludes this article with a discussion of the results.

II. Related Works

A. Geospatial Object Detection

Object detection aims at simultaneously localizing and classifying the instances in a given image and is considered one of the fundamental and challenging tasks in computer vision. The currently mainstream deep-learning-based methods can be

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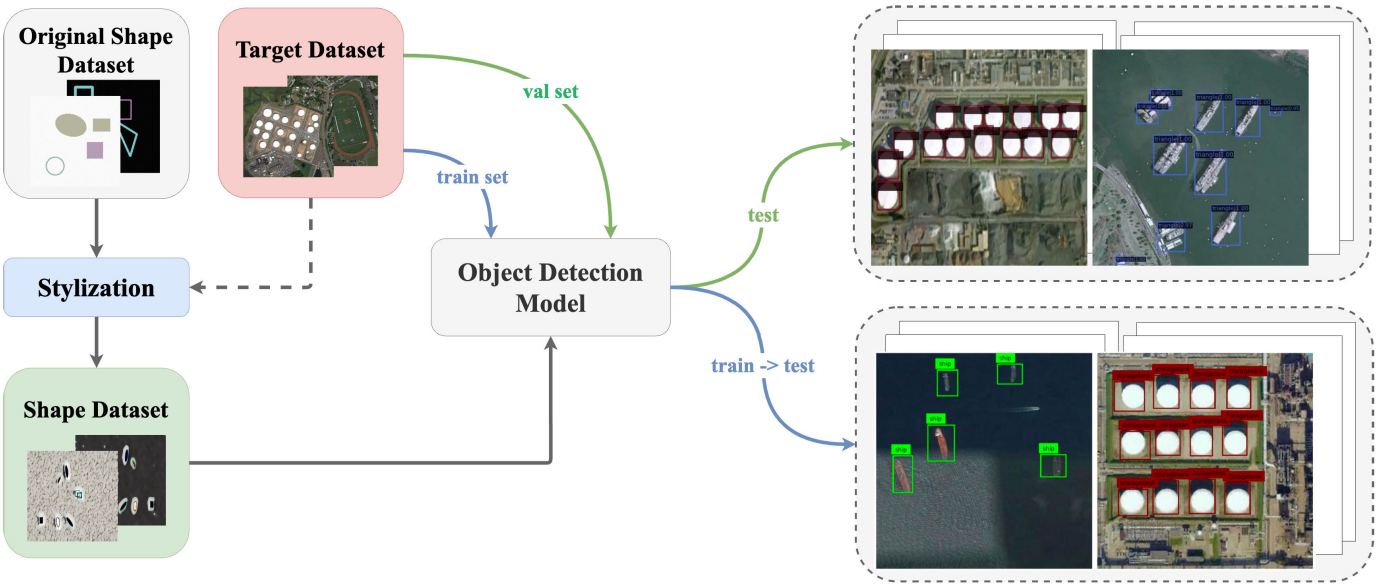


Fig. 2. Framework of the SPCG data augmentation method. This approach begins by generating a basic SD, which serves to train the model in acquiring shape priors. These priors are instrumental in enabling precise target location perception within remote sensing images, as detectors trained on this SD are capable of forming significant intermediate representations, even without training on the actual target dataset. Then, the model with shape priors further trains on the target dataset. This step can transfer generalized shape knowledge to specific tasks, enhancing the model's ability to recognize and interpret distinct target objects.

roughly grouped into one-stage [22], [23], [24], [25] and two-stage detectors [26], [27], [28], [29]. The difference between the two lies in the subsequent usage of high-level features, which are extracted by general backbones [30], [31] possibly with FPN [32]. The one-stage detectors directly predict classification and localization results based on the extracted high-level features. In contrast, the two-stage detectors first put the features into a region proposal network (RPN) to generate a set of object proposals and then leverage them to produce the results. Besides, another criterion has arisen in recent years that divides detectors into anchor-based [23], [25], [26] and anchor-free ones [24], [33], [34], according to whether predefined sliding windows (i.e., anchors) are used or not. More recently, transformer-based detectors like DETR [35], [36] have emerged, offering a different paradigm that eliminates the need for anchors and proposes objects through global reasoning, enriching the variety of techniques available for object detection.

The complex data characteristics of remote sensing images pose great challenges to the task of remote sensing object detection [1], [2], [21]. Specifically, the instances in remote sensing images often exhibit large aspect ratios, inconsistent orientations [1], and varying scales [2]. Furthermore, the current high-resolution remote sensing images are notoriously stuck with high intraclass variation and low interclass distinction, let alone some unwanted imaging noises [26]. In the past few years, many remote sensing object detection methods have been proposed with the data characteristics taken into consideration. For example, Liu et al. [37] focused on the scale problem and proposed a special module to fuse the context in multiscale features. Xu et al. [38] imputed the lagged detection performance for the instances with large aspect ratios and varying scales to the problem of feature misalignment and proposed a feature-aligned detection method. Cheng et al. [1] put forward a feature enhancement network and a context encoding loss to promote the separability for interclass instances.

B. Data Augmentation

At present, prevailing methods for augmenting remote sensing data fall into three primary categories: global transformations, element synthesis, and conditional generation. Global transformation techniques in data augmentation include random transformation, image erasing, and image mixing. Mixup [7] blends two images from the training set by linearly combining their pixel values and labels, creating images with blurred classification boundaries for improved model generalization. CutMix [6] enhances target localization by replacing parts of an image with patches from another, adjusting the labels based on the pixel contribution of each image; however, it is more suited for classification than detection tasks. The CopyPaste method [9] boosts detector performance by randomly pasting objects into images. Following these classic techniques, a series of advanced Mix-based methods have emerged, including Co-mixup [39], AlignMixup [40], TokenMix [41], and MixPro [42], each offering novel variations and improvements on the original mixing concepts.

The element synthesis method leverages elements from various remote sensing images, such as radar, satellite, and aerial photographs, to create realistic images. Drawing on the physical properties of these images, data reconstruction is performed. Ding et al. [10] enhanced SAR images by using attribute scattering centers (ASCs), minimizing noise and background interference. Nonetheless, the process of image reconstruction is computationally intensive, and its accuracy can be limited.

Conditional generation methods utilize deep generative models to enrich remote sensing data. These generative

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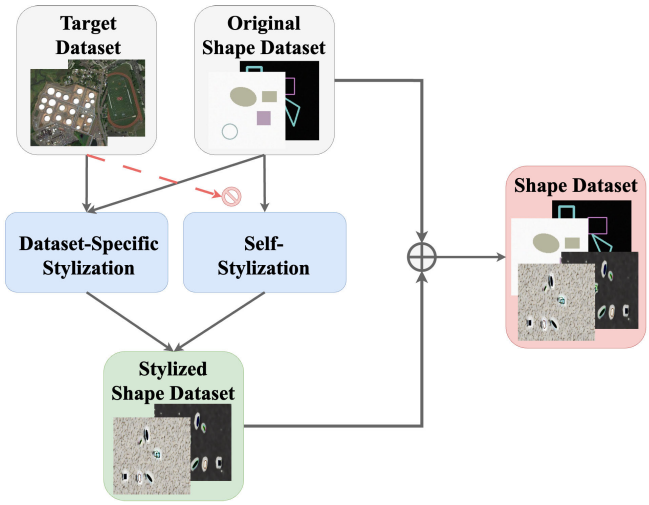


Fig. 3. SD is composed of two parts: the O-SD and the S-SD. Depending on the availability of the target dataset, we employ two different methods for stylization—self-stylization and dataset-specific stylization.

models can be classified into likelihood-based models [43] and implicit generative models [11], [12]. The former, such as autoregressive models [44], [45], flow models [46], and variational auto-encoders (VAE) [47], learn the probability distribution of the data using the maximum likelihood criterion. In contrast, the latter, which includes GANs [11] and DDIMs [14], lacks an explicit representation of the probability distribution. Specifically, GANs generate realistic images through adversarial training between the generator and discriminator networks, while DDIMs sample image results iteratively from noise distribution. Among the GAN variants, Cycle-GAN [13] has gained prominence for its ability to perform unpaired image-to-image translation, facilitating the generation of remote sensing images in different styles without needing corresponding pairs in the target domain. Additionally, recent developments like IT-GAN [48] and BigDatasetGAN [49] further advance the field by offering improved training stability and the ability to generate high-quality images from large-scale datasets, respectively. While conditional generation methods offer benefits, they demand a large volume of training data and come with high computational costs. Our shape-prior-based augmentation method also falls under the generative category. In stark contrast, our method not only incurs low computational costs but also offers a generative data augmentation method that requires no training data.

III. Method

The proposed SPGC data augmentation method first trains the model on the SD and then trains on the target dataset. As shown in Fig. 3, the SD consists of the original SD (O-SD) and the stylized SD (S-SD). Their implementation details are as follows.

A. Original SD

1) Original SD: The O-SD contains various shapes that can be easily generated by code. In our setting, we use Opencv-Python to generate rectangles, circles, triangles and ellipses, as they are one of the most basic shapes that can compose various complex shapes, such as trapezoids can be composed of rectangles and triangles. We also verified the situation of different shape categories in the ablation experiment. In order to simulate the diversity of shape in the real world to the greatest extent, the number, position, size, color, and rotation angle of shapes in each image are completely random. The code for shape data generation is simple and easily modifiable, so these generation-related parameters are of course fully controllable if needed. In a composite image, for each basic shape object, its category and bounding box can be obtained from the previous random generating process. By combining all the required coordinates and categories, the label of this composite image is generated. Given an image I in O-SD, the metadata consisting of categories C and bounding boxes B can be formulated as a labeled set $L_I = \{(c_i, b_i) \mid c_i \in C, b_i \in B\}$. The generated shape image of O-SD is shown in the first column of Fig. 4.

B. Self-Stylized SD

1) Self-Stylized SD: On the basis of the O-SD, we generate a S-SD using the method of self-style transfer when the target dataset is unseen. Specifically, we use the O-SD itself as the style source to stylize the O-SD, which we call self-stylized (SS) since the style image and the content image are the same. We adopt the AdaIN style transfer [50] in our experiments

$$F_{\text{AdaIN}}(O_i, O_j) = \sigma(O_j)\left(\frac{O_i - \mu(O_i)}{\sigma(O_i)}\right) + \mu(O_j) \quad O_i, \quad O_j \in \mathcal{O} \tag{1}$$

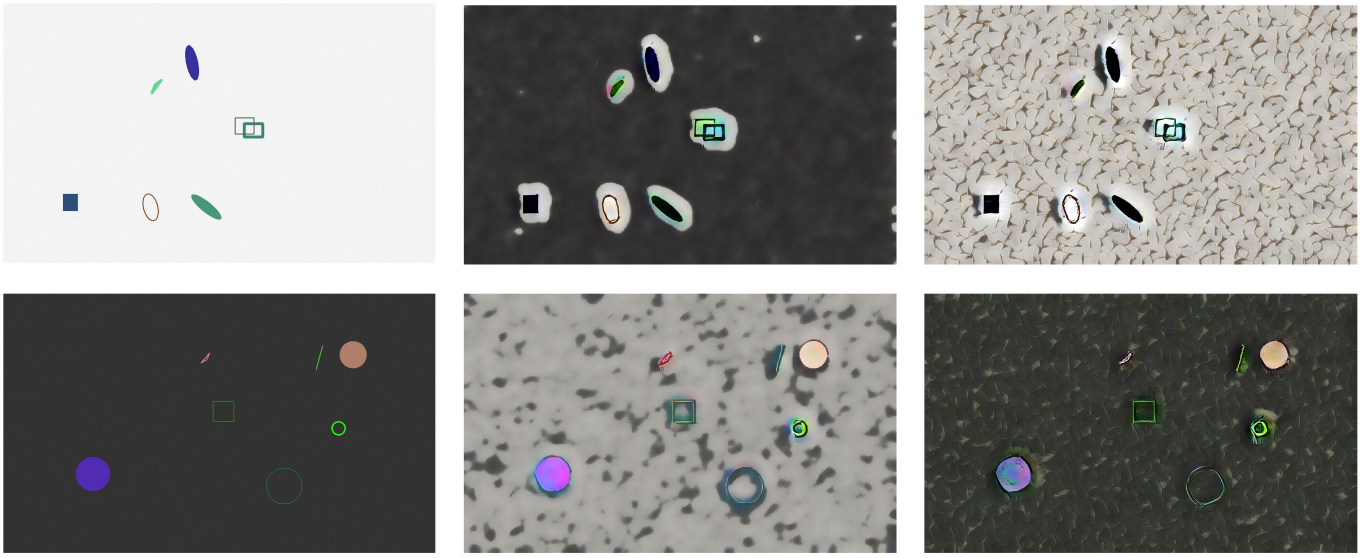
where O represents the O-SD, $\sigma(O)$ represents the image mean, and $\mu(O)$ represents the image variance.

Therefore, our main purpose of using stylization is not to achieve true style transfer, but to obtain irregular edges and complex textures. The data augmentation approach from self-stylization is used to simulate real-world situations where most shapes are not standard shapes and the variety of textures and backgrounds of shapes. For example, in real life, we also regard a rectangle with rounded corners as a rectangle. We use image information entropy to characterize the randomness of images as follows:

$$H_{2D}(\mathcal{I}) = -\sum_{i=0}^{255} P(g_i) \log_2 P(g_i) \tag{2}$$

where $I \in O \cup O_{\text{SD}}$ is an image and $P(g_i)$ is the probability of pixel value i appearing in the image. As can be seen from Fig. 5, the entropy H_{2D} of the image increases significantly after the stylization operation. This demonstrates that the self-stylization operation successfully introduces a large amount of uncertainty and randomness into the original image dataset, which better simulates real-world situations. More importantly, the style transfer changes the image texture significantly, but hardly changes the overall shape and position of the image as shown in the second column Fig. 4. So for the S-SD, we can use the same annotations as the O-SD with no doubt.

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(a) (b) (c)

Fig. 4. Comparison of (a) OS images, (b) SS images, and (c) DS images. In self-stylization, the data enhancement process dramatically alters the texture within and around the shape while also introducing irregularities to its edges. This mimics real-world shape irregularities and randomness, significantly improving the robustness of the shape detector. On the other hand, dataset-specific stylization not only adds randomness to the images but also aligns the distribution of the SD more closely with that of the target dataset, leading to further improvements in object detection performance on the target dataset.

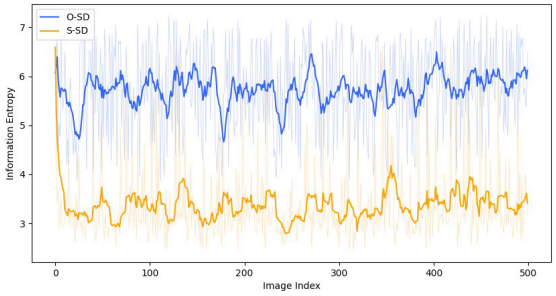


Fig. 5. Information entropy of each image in O-SD and S-SD. The result is the average information entropy computed from five random samplings of 500 images each.

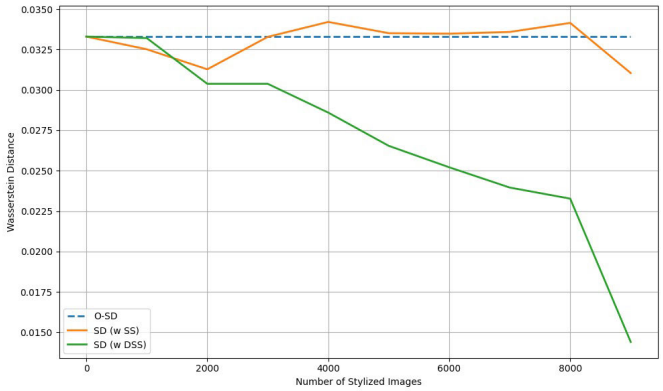


Fig. 6. Wasserstein distance comparison between O-SD, SD with self-stylization (w SS), and SD with dataset-specific stylization (w DSS). For every additional 1000 stylized images generated, the Wasserstein distance between the SD and the target dataset is computed and visualized.

C. Dataset-Specific Stylized SD

1) Dataset-Specific Stylized SD: When training data is available, dataset-specific stylization is applied. Specifically, we use the target dataset as the style source to stylize the O-SD. Like Self-stylization, we also adopt the AdaIN style transfer [50] in our experiments

$$F_{\text{AdaIN}}(O_i, T_j) = \sigma(T_j)\left(\frac{O_i - \mu(O_i)}{\sigma(O_i)}\right) + \mu(T_j) \qquad O_i \in \mathcal{O}, \quad T_j \in \mathcal{T} \tag{3}$$

where O represents the O-SD and T represents the target remote sensing dataset. Going a step further than self-stylization, dataset-specific stylization aims not only to introduce random variations but also to bring the SD closer to the distribution of the target training dataset. We use the Wasserstein distance [12] to measure the distance between two dataset distributions

$$W(\mathcal{O}_{\text{DS-S}}, \mathcal{T}) = \inf_{\gamma \in \Pi(\mathcal{O}_{\text{DS-S}}, \mathcal{T})} \int_{\mathcal{O}_{\text{DS-S}} \times \mathcal{T}} d(x, y) d\gamma(x, y) \tag{4}$$

where $\mathcal{O}_{\text{DS-S}}$ is a collection of images generated by dataset-specific stylization, $\Pi(\mathcal{O}_{\text{DS-S}}, \mathcal{T})$ is the set of all joint distributions between these two sets. As can be seen from Fig. 6, as the number of dataset-specific stylized (DSS) images increases, the distance between the two datasets decreases gradually. Utilizing t-SNE for dimensionality reduction, we depict the data distribution from various datasets (i.e., target dataset, O-SD, and S-SD employing DSS) via scatter plots, as illustrated in Fig. 7. As observed from Fig. 7, after the application of dataset-specific stylization, the distribution of the S-SD data tends to diffuse toward the target dataset compared to its original position in O-SD. Therefore, DSS successfully shortens the distance between the SD and the distribution of the target dataset. It can also be seen from the third column Fig. 4 that the background and texture of the shape data after DSS are closer to the target remote sensing dataset. Compared with generative methods such as GAN, only a small amount of computing power can be used to obtain data close to the real distribution.

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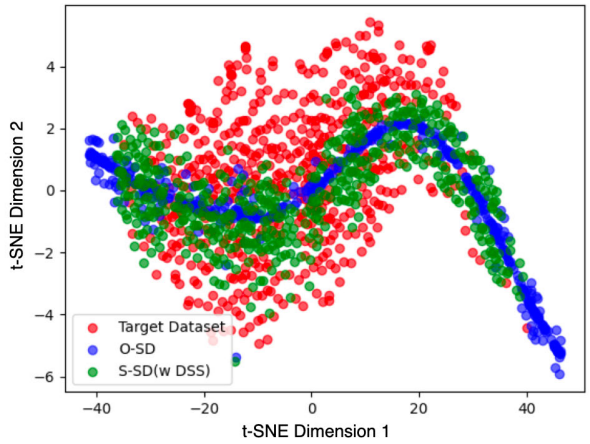


Fig. 7. t-SNE visualization presented samples from three datasets: the target dataset, the O-SD, and the S-SD that utilizes dataset-specific stylization (w DSS). For this representation, 500 samples (images) were randomly selected from each of the datasets. In this experiment, DIOR is used as the target dataset.

IV. Experiment

In this section, we first introduce the dataset and evaluation metrics. Then, the results on mainstream object detection datasets are shown. Finally, ablation experiments and visualizations were performed.

A. Dataset

To evaluate the proposed structure, extensive experiments are conducted on two coarse-grained datasets: NWPU VHR-10 [1], DIOR [2], and one fine-grained dataset: FAIR1M [21].

1) NWPU VHR-10: This dataset comprises 800 images. The image resolution ranges from a maximum of 0.08 m to a minimum of 2 m. Objects with different scales in this dataset are categorized into ten classes. It consists of a positive image set including 650 images and a negative image set including 150 images over ten object categories with different image sizes (i.e., from 533 × 597 to 1728 × 1028 pixels).

2) DIOR: DIOR is a popular public large-scale dataset for object detection in remote sensing images. The size of the images in DIOR is 800 × 800 pixels and the spatial resolutions range from 0.5 to 30 m. It consists of 23 463 images and 192 472 instances, covering 20 object classes. We follow the official strategy to split the dataset. Specifically, 5862 images are used for training, the validation set contains 5863 images, and the remaining 11 738 images for the test set.

3) FAIR1M: FAIR1M contains 42 796 images which have a ranging resolution from 0.3 to 0.8 m, including 1.02 million objects. All the images are captured by Gaofen satellites or collected from Google Earth. It has a fine-grained category design. All the objects are divided into 5 classes and 37 subclasses, containing 11 classes for airplanes, 9 classes for ships, 10 classes for vehicles, 4 for courts, and 3 for roads. The images for training and testing are cropped to 512 × 512. The degraded images are with the size of 64 × 64, 32 × 32, and 16 × 16, and the resolution changes from 1.2 to 3.2 m, from 2.4 to 6.4 m, and from 4.8 to 12.8 m.

B. Experiment Settings

1) Evaluate Metrics: We evaluate object detection results with the standard COCO protocol. We report the average precision (AP) and average recall (AR) on validation split, which is averaged over ten intersection-over-union (IoU) thresholds evenly spaced between 0.5 and 0.95.

2) Shape Dataset: We generate the SD according to the process in Section III. In the setting of all our experiments, 9000 images are generated with a total of 61 198 shapes, and 1 to 15 shapes randomly present in each image. For the self-stylization process, we adopt the AdaIN style transfer [50] and set the stylization coefficient α to 0.2.

3) Implement: For the implementation of PnP DETR and Sparse DETR, we adhered to their official implementation code [54], [55]. For all YOLO-based methods (including YOLOv5, YOLOv7, YOLOv8, and YOLOX), our implementation is based on MMYOLO [57]. Other detectors such as Faster RCNN, Mask RCNN, CenterNet, and Deformable DETR, are implemented using MMDetection [58]. In line with MMDetection's default settings, we employed the SGD optimizer for CNN-based models and the AdamW optimizer for transformer-based models. Notably, the SD, being relatively simple, did not require customized training strategies or parameters. All experiments were conducted using Pytorch on Ubuntu 20.04, with NVIDIA RTX 3090 GPUs and CUDA 11.3.

For the GAN-based methods in our experiments, we adhered to original model structures and training strategies [13], [48], [49]. Specifically, for CycleGAN, training was from scratch, with a learning rate of 0.0002 for the first 100 epochs, followed by linear decay, and the discriminator's learning rate was adjusted to half of the generator's. For IT-GAN, we utilized ConvNet and ResNet18, applying SGD optimization over 200 epochs with a two-phase learning rate. In BigDataset GAN, we implemented DeepLab-v3 with ResNet-50, using SGD with a learning rate of 0.01, momentum, and weight decay, alongside polynomial learning rate decay and standard augmentations. For all generated-based methods, we generated the same number of images as those in the SD.

C. Experiments on Diverse Object Detectors

In our evaluation of mainstream object detectors, we have chosen a wide array of models to demonstrate the effectiveness of our method. This includes a selection of one-stage detectors such as YOLOV5 [22] and YOLOv7 [51], as well as the two-stage detectors Faster RCNN and Mask RCNN [29]. Additionally, we have considered anchor-free detectors like CenterNet [34] and YOLOX [53], and transformer-based detectors including deformable DETR [35] and Sparse DETR [55]. These choices encompass a range of CNN-based two-stage, one-stage, anchor-free, and transformer-based methods, offering an extensive overview of the current state-of-the-art in object detection.

As shown in Table I, our SPGC method consistently achieves significant improvements across various detection models on the NWPU VHR-10 dataset, surpassing the baseline. Particularly noteworthy is the performance boost

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TABLE I

EVALUATING VARIOUS ARCHITECTURES FOR OBJECT DETECTION ON THE NWPU VHR-10 DATASET. BASELINE IS THE OBJECT DETECTORS WITH BASIC DATA AUGMENTATION (RANDOM FLIP AND CROP). FOR CLARITY, THE FOLLOWING ABBREVIATIONS ARE USED IN THIS TABLE: FASTER RCNN AS F-RCNN, MASK RCNN AS M-RCNN, DEFORMABLE DETR AS D-DETR, PNP DETR AS P-DETR, AND SPARSE DETR AS S-DETR

Method	One-stage			Two-stage		Anchor-free		Transformer-based		
	YOLOv5 [52]	YOLOv7 [51]	YOLOv8 [52]	F-RCNN [26]	M-RCNN [29]	CenterNet [34]	YOLOX [53]	D-DETR [35]	P-DETR [54]	S-DETR [55]
Baseline	45.2	55.8	59.2	50.8	52.5	48.9	59.1	4.3	48.5	50.9
SPGC (SS)	47.5	58.1	60.7	56.7	58.7	53.4	63.3	52.6	50.6	57.0
SPGC (DSS)	**49.1**	**59.3**	**61.1**	**57.5**	**60.1**	**55.1**	**65.0**	**54.8**	**52.9**	**58.3**

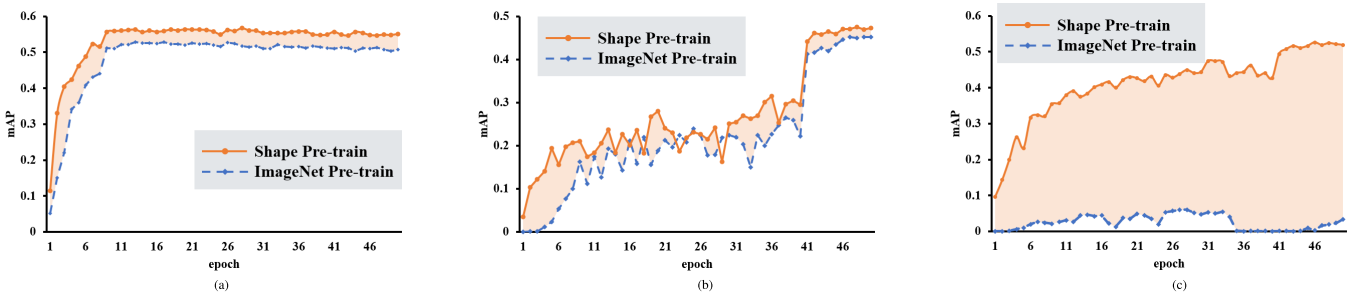


Fig. 8. SPGC data augmentation improves the detection network of various structures. (a) Faster RCNN. (b) YOLOV5. (c) Deformable DETR.

TABLE II

COMPARISON WITH THE STATE-OF-THE-ARTS DATA AUGMENTATION METHOD ON DIOR AND FAIR1M. BASELINE IS THE OBJECT DETECTORS WITH BASIC DATA AUGMENTATION (RANDOM FLIP AND CROP). IN OUR SPGC METHOD, WE ADDITIONALLY FREEZE THE FIRST TRAINABLE LAYER OF THE BACKBONE (RESNET 50) DURING TRAINING ON THE TARGET DATASET

Method	Pub.	DIOR	FAIR1M
Baseline	-	68.3	31.5
RandAug [58]	CVPRW2021	68.8	33.1
Mixup [7]	ICLR2018	69.7	33.6
CutMix [6]	ICCV2019	67.6	30.0
CopyPaste [9]	CVPR2021	70.1	33.5
Co-mixup [39]	ICLR2021	71.6	34.1
AlignMixup [40]	CVPR2022	72.3	34.9
TokenMix [41]	ECCV2022	71.0	33.7
MixPro [42]	ICLR2023	71.2	34.5
Cycle-GAN [13]	ICCV2017	73.3	36.5
IT-GAN [48]	NIPSW2022	75.2	37.9
BigDatasetGAN [49]	CVPR2022	74.8	38.6
SPGC(SS)	-	75.0	38.3
SPGC(DSS)	-	**76.3**	**40.2**

observed in transformer-based detectors like Deformable DETR, which had initially shown suboptimal performance on this dataset, an issue likely stemming from its relatively small size and the presence of smaller objects [35]. Further, as illustrated in Fig. 8, SPGC provides substantial performance enhancements compared to mainstream object detectors using basic data augmentation techniques such as random flipping and cropping. The integration of SPGC not only consistently improves performance throughout the training process but also contributes to faster convergence of the detection network, demonstrating its adaptability and efficacy across a broad spectrum of advanced detection frameworks.

D. Experiments on DIOR and FAIR1M Datasets

We further evaluate our proposed SPGC method by benchmarking it against state-of-the-art data augmentation techniques on two popular datasets—DIOR, a large-scale dataset, and FAIR1M, a fine-grained dataset. Our method is compared with image transformation-based approaches such as Mixup [7], CutMix [6], CopyPaste [9], Co-mixup [39], AlignMixup [40], TokenMix [41], and MixPro [42]. Additionally, it is compared with deep learning-based generative methods, such as Cycle-GAN [13], IT-GAN [48], and BigDatasetGAN [49]. For implementation details and training specifics of the GAN-based methods, please refer to Section IV-B.

As depicted in Table II, SPGC achieves a discernible increase in the mean AP (mAP) metric. For example, when compared to the GAN-based Method, SPGC exhibits a performance gain of 6.8% in mAP on the DIOR dataset and 9.2% on the FAIR1M dataset. Overall, SPGC consistently surpasses other methods on both DIOR and FAIR1M datasets, thereby substantiating its efficacy in handling complex remote sensing images, including large-scale and fine-grained data.

E. Visualization

1) What Exactly is Shape Detection?: To solve this problem, we made numerous visualizations of the results of the shape detector (see Fig. 9). The results show that the shape detection results are similar to the ground truth boxes in many images. And for objects with distinctive shape characteristics, accurate categories can be obtained. We also conducted a **class-agnostic** experiment to quantify the effect of the shape detector (see Table III), showing that even without any training data, the shape-based detectors achieve a higher recall.

F. Comparison With the Pretraining Methods

Given the procedural similarities between our two-stage training approach and conventional pretraining methods, we conducted comparisons with mainstream pretraining approaches to further explore the relationship between

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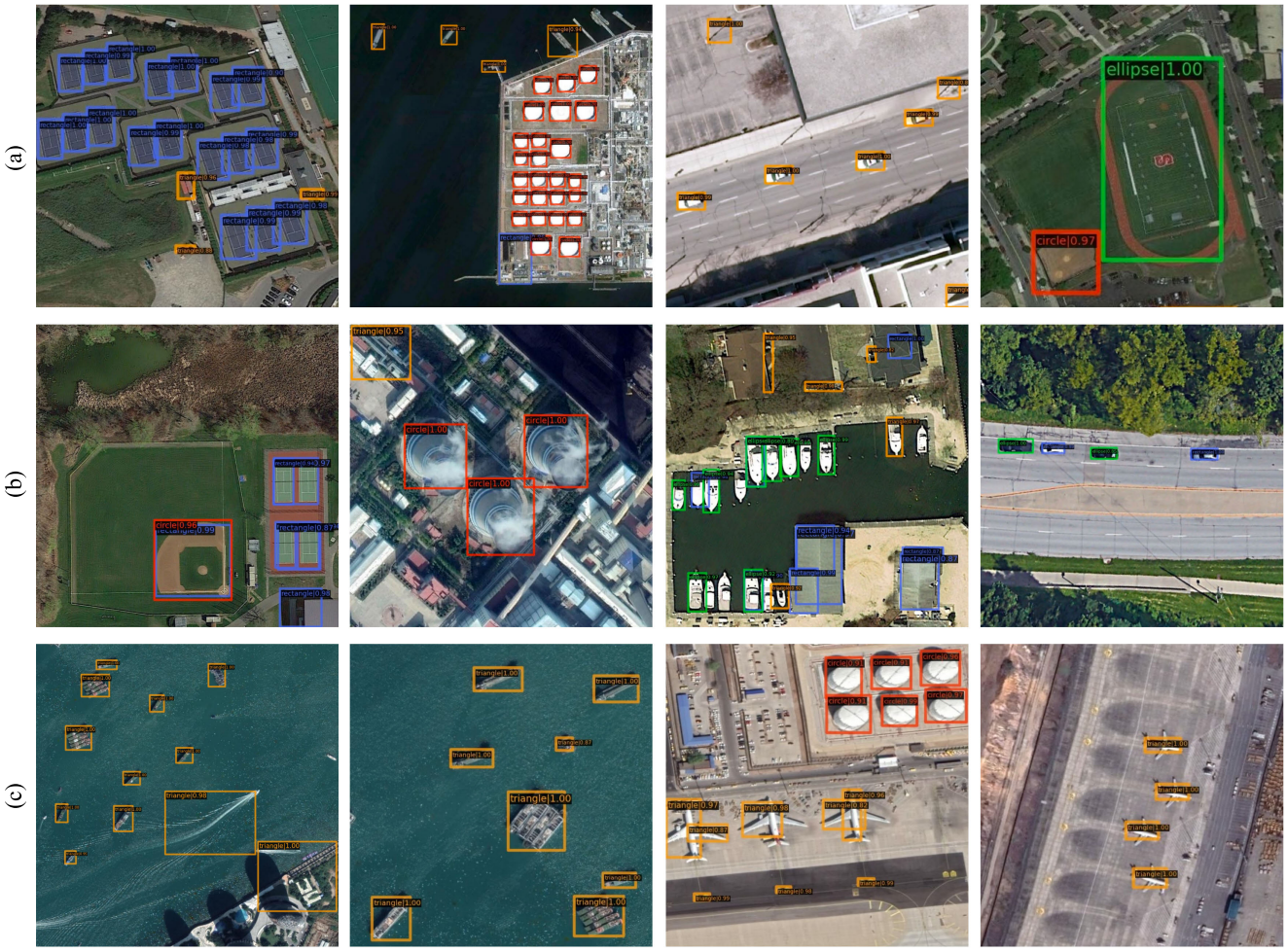


Fig. 9. Shape detection visualization. We present the results of the detector, trained on the SD, and test on mainstream remote sensing datasets (a) NWPU VHR-10, (b) DIOR, and (c) FAIR1M. The blue, red, yellow, and green bounding boxes in the result figures represent rectangles, circles, triangles, and ellipses, respectively.

TABLE III
CLASS AGNOSTIC OBJECT DETECTION. MODEL TRAINED ON SD IS TERMED AS SPGC

	Training Data	AR_s	AR_m	AR_L
Selective Search	-	0.1	3.5	10.7
RPN	✓	0.22	0.62	0.57
RPN SPGC.	-	0.39	0.37	0.18
D-DETR SPGC.	-	0.13	0.19	0.47
YOLOV5 SPGC.	-	0.53	0.24	0.53
FRCN SPGC.	-	0.40	0.29	0.54

shape knowledge and the "general knowledge" learned through large datasets. Specifically, we compare our method with well-known pretraining approaches like ImageNet Pretrain [59] and MoCo [60]. Furthermore, we integrated SPGC with pretraining methods designed explicitly for remote sensing optical images, including RSP [61] and SSL4EO-L [62].

As shown in Table IV, our experimental results reveal that our method significantly outperforms the ImageNet pre-training and self-supervised pretraining methods MoCo [60] on natural images, even without using remote sensing data for stylization. This indicates that shape features are more beneficial for remote sensing data compared to features extracted from multiscale natural images. When

TABLE IV
COMPARISON WITH THE PRETRAINING METHODS. "SPGC(DSS)" DENOTES APPLYING OUR METHOD FROM SCRATCH. "+ SPGC(DSS)" INDICATES INITIALIZING MODEL PARAMETERS WITH THE PRETRAINED WEIGHTS OF THE CORRESPONDING METHODS BEFORE APPLYING OUR SPGC METHOD

Method	Pub.	NWPU VHR-10
ImageNet Pre. [59]	-	50.8
MoCo v2 [60]	CVPR 2020	52.3
SPGC(SS)	-	56.7
SPGC(DSS)	-	**57.5**
RSP [61]	TGRS 2022	61.4
+ SPGC(DSS)	-	**63.0**
SSL4EO-L [62]	NIPS 2023	60.8
+ SPGC(DSS)	-	**61.3**

applied on top of models pretrained on large-scale remote sensing datasets, our method still enhances performance in remote sensing image object detection. This highlights the vital role of shape and location priors provided by our method in remote sensing image object detection.

G. Ablation Study

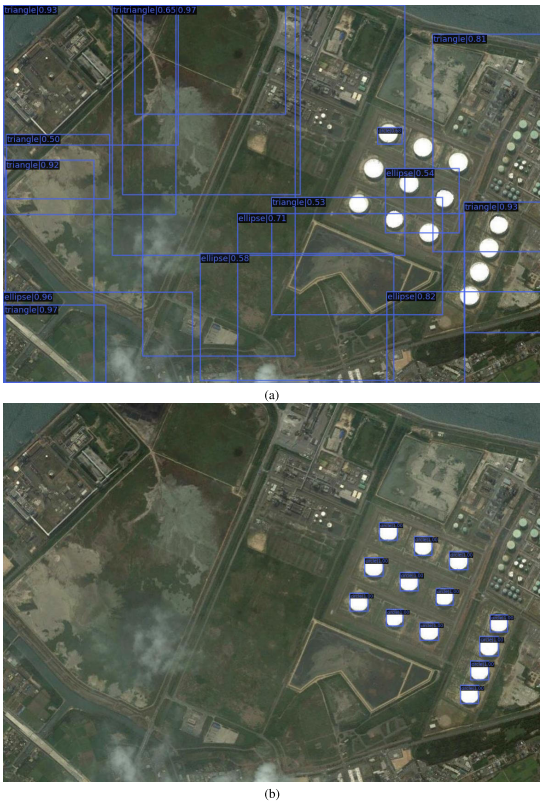
We conduct ablation experiments on NWPU VHR-10 to show how our components work. Experimental settings are the same as Section IV-B.

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TABLE V

ABLATION STUDY ON STYLIZATION. THE COMPARISON ON NWPU VHR-10 AMONG O-SD, S-SD, AND SD. THE SD CONSISTS OF O-SD AND S-SD. SPECIFICALLY, SD(DSS) REPRESENTS A MIXTURE OF O-SD AND S-SD(DSS)

Dataset	Stylization	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
O-SD	-	39.8	66.2	51.0	3.4	43.2	31.1
S-SD(SS)	✓	53.7	86.2	63.1	6.4	54.8	50.5
S-SD(DSS)	✓	56.1	86.5	63.4	7.0	56.1	51.6
SD(DSS)	✓	**56.7**	**88.3**	**63.2**	**6.7**	**57.1**	**52.0**



(a)

(b)

Fig. 10. Ablation study on SS augmentation. The model trained on stylized data has fewer and more accurate detection results, almost the same as the ground truths. But the results trained on O-SD are messy and many, and most of the targets are missed. (a) Training on O-SD. (b) Training on S-SD.

1) Stylization: To illustrate the importance of the Stylization method, we compare the O-SD with the S-SD and SD. Table V and Fig. 10 show that SS augmentation for SD is the main reason why our method is effective.

2) Quantity and Category of Shapes: In our ablation experiments, we specifically investigate the influence of two key parameters on the performance of our proposed SPGC framework: the quantity of shape images q and the number of shape categories c. All other experimental settings remain constant to Section IV-B except for q and c.

As depicted in Fig. 11(a), we observe a noteworthy pattern in the model's performance with respect to q. When q is too small, the benefits to the detection models from the SD are minimal. As q increases, the object detector experiences incremental gains in terms of mAP. However, the performance begins to plateau when q surpasses a certain threshold (approximately 10 000 in our case). One plausible explanation is that a sample size around 10 000 is sufficient for the detector

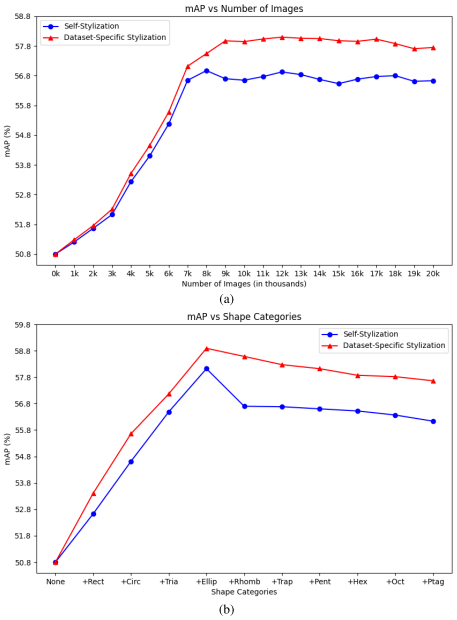


Fig. 11. Ablation study on number of images q and shape categories c. The detection model is Faster RCNN and tests on the NWPU VHR-10 dataset. (a) Ablation study on the number of images. (b) Ablation study on shape categories.

to learn adequate shape representations. Introducing more samples beyond this point may lead to overfitting without offering further improvements in the detector's understanding of shapes.

For the variable c, the number of shape categories, we experiment with up to ten different shapes. The shapes are introduced incrementally in the following sequence: rectangle, circle, triangle, ellipse, rhombus, trapezoid, pentagon, hexagon, octagon, and pentagram. As shown in Fig. 11(b), the trends we observe for c largely mirror those for q. This could be attributed to two factors.

1) Most shapes can be approximated or represented by a combination of the initial four basic shapes. For instance, a trapezoid can be constructed from rectangles and triangles.
2) The shapes in real-world remote sensing images are seldom perfectly standardized. For example, an oil tank might be detected as an octagon, a circle, or an ellipse. Learning too many shape categories could increase the detector's uncertainty and lead to phenomena such as multiple bounding boxes surrounding a single object.

V. CONCLUSION

In this work, we proposed a novel data augmentation technique, SPGC, to address data scarcity and variability issues in remote sensing object detection. Utilizing shape bias from human cognition as inspiration, SPGC comprises two elements: original shape data generation and stylization. Experimental validation on leading datasets, including NWPU VHR-10, DIOR, and FAIR1M, confirms SPGC's effectiveness in significantly enhancing model performance. Notably, our approach is both computationally efficient and easily adaptable

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to mainstream object detection architectures. Future work may explore SPGC's applicability in other remote sensing domains.

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Yalun Dai received the B.E. degree in communication engineering from the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing, China, in 2022. He is currently pursuing the master's degree with the School of Electrical and Electronic Engineering.

His research interests include deep learning, computer vision, and artificial intelligence.



Fei Ma (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic and information engineering from Beihang University, Beijing, China, in 2013, 2016, and 2020, respectively.

From 2017 to 2018, he was a Research Fellow with the Department of Electrical Engineering, McGill University, Montreal, QC, Canada. He is currently an Associate Professor with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing.

His research interests include synthetic aperture radar (SAR) image processing, machine learning, artificial intelligence, and target detection.



Wei Hu received the B.E. and M.S. degrees in computer science and technology from Dalian University of Science and Technology, Dalian, China, in 1999 and 2002, respectively, and the Ph.D. degree in computer science and technology from Tsinghua University, Beijing, China, in 2006.

He is currently an Associate Professor of computer science and technology with Beijing University of Chemical Technology, Beijing. He has published more than 60 articles, including CVPR, ICLR, ICCV, IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, ACM SI3D, and Pacific Graphics. His research interests include deep learning training strategy, AI music analysis, face recognition, and real-time graphics rendering.



Fan Zhang (Senior Member, IEEE) received the B.E. degree in communication engineering from the Civil Aviation University of China, Tianjin, China, in 2002, the M.S. degree in signal and information processing from Beihang University, Beijing, China, in 2005, and the Ph.D. degree in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2008.

From December 2013 to January 2015, he was a Visiting Scholar working with Prof. Martin D. F. Wong at the Coordinated Science Laboratory, University of Illinois Urbana–Champaign, Urbana, IL, USA. He is currently a Full Professor of electronic and information engineering with Beijing University of Chemical Technology, Beijing. He has published more than 100 peer-reviewed articles, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *Photogrammetric Engineering and Remote Sensing*, CVPR, and ICCV. His research interests include synthetic aperture radar (SAR) system simulation, SAR image interpretation, and artificial intelligence.

Dr. Zhang is an Editorial Board Member of *Journal of Radars*.

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